25901736 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCMBG100 - PERSPECTIVES OF MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. How do Streptococci differ from Staphylococci in their arrangement? [U] / [CO1]
2. Explain naked and enveloped viruses. [U] / [CO1]
3. Name two common asexual reproductive structures in fungi. [K] / [CO1]
4. What is endospore? [K] / [CO1]
5. List out the basic components in a bacteriological culture media [K] / [CO2]
6. Name three special media. [K] / [CO2]
7. Name any two physical factors affecting growth of bacteria [K] / [CO2]
8. Classify bacteria based on electron sources. [U] / [CO2]
9. Define microaerophiles. Give examples. [K] / [CO2]
10. What is the purpose of turbidimetric method ? [K] / [CO3]

11. Why does the growth rate decline in the stationary phase? [K] / [CO3]

12. Write down the use of the coulter counter. [U] / [CO3]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

13. Differentiate between psychrophiles and thermophiles with examples. [U] / [CO2]

14. What are growth factors, types and importance? Give examples. [K] / [CO2]

15. Contrast between Budding and Fragmentation. [U] / [CO3]

16. Compare and contrast the pour plate, spread plate and membrane filteration methods for bacterial enumeration. [U] / [CO3]

17. Summarize the non living characteristics of viruses. [U] / [CO1]

18. Explain budding in yeast. [U] / [CO1]

[5x4 = 20]

Part C

Long Essay Question

Answer any **1** questions

Each question carries **10** marks

19. Define total count method and explain the methods used to determine total bacterial count. [K] / [CO3]

20. What is culture media? Explain different types of culture media based on consistency with examples. [U] / [CO2]

[10x1 = 10]